

DASC7606

Deep Learning

Vision Applications:

Localization and Detection

Dr Bethany Chan

Professor Francis Chin

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ResNet revisit

Vanishing Gradient Problem

Forward feeding (from i to $i+1$) $\rightarrow$

- $s_i = W_i x_i + b_i$
- $x_{i+1} = \sigma(s_i)$ where $\sigma = \text{sigmoid}$

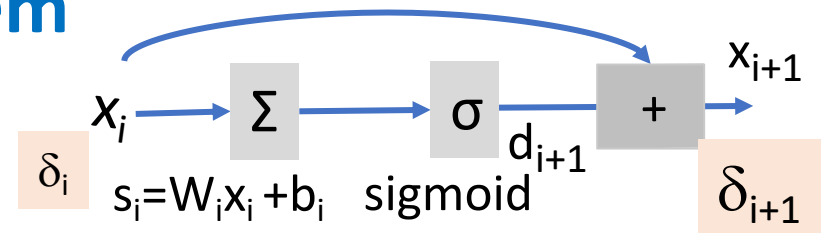
BP to find the value of $\delta_i = \partial J / \partial x_i$ (from $\delta_{i+1} = \partial J / \partial x_{i+1}$) $\leftarrow$

- $\delta_i = \partial J / \partial x_i = (\partial J / \partial x_{i+1}) (\partial x_{i+1} / \partial x_i) = \delta_{i+1} (\partial x_{i+1} / \partial s_i) (\partial s_i / \partial x_i)$
 $= \delta_{i+1} \sigma(s_i) (1 - \sigma(s_i)) W_i$

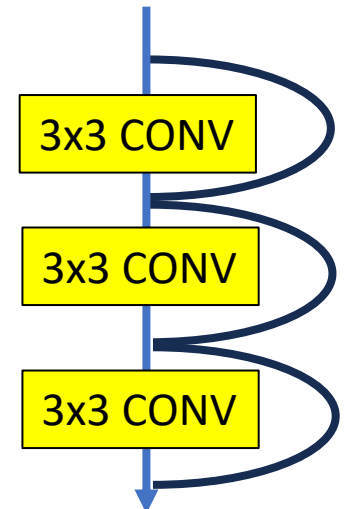
With skip path

- $x_{i+1} = x_i + d_{i+1}$; $d_{i+1} = \sigma(s_i)$
- $\delta_i = \partial J / \partial x_i = (\partial J / \partial x_{i+1}) (\partial x_{i+1} / \partial x_i) = \delta_{i+1} (1 + \partial d_{i+1} / \partial x_i)$
 $= \delta_{i+1} (1 + \sigma(s_i) (1 - \sigma(s_i)) W_i)$

value won't decrease



$$\sigma(s_i) \leq 0.25$$



Various filter sizes (GoogLeNet)

By skipping some of the 3x3 CONV,

We can have effectively 3X3, 5x5 or 7x7 filters

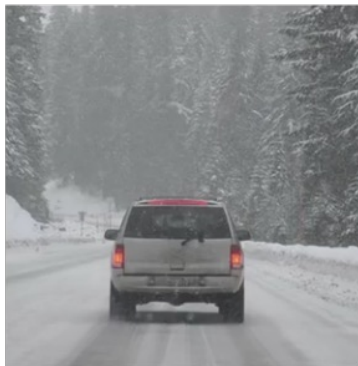
Various Computer Vision Tasks

- Visualizing and Understanding CNN
- **Object Localization**
- Object Detection
 - Sliding windows
 - R-CNN, Fast R-CNN, Faster R-CNN, YOLO
- Semantic Segmentation – Mask R-CNN

Object Localization

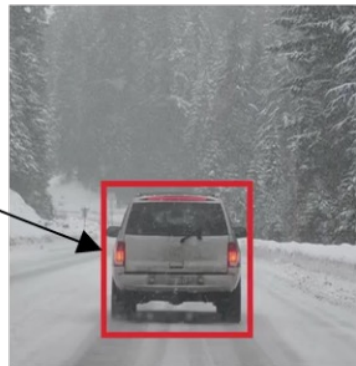
- **Image classification** – identify an object in a picture
- **Classification with localization**
 - identify an object with a bounding box
- **Detection** - detect and localize all objects (usually >1)
e.g., cars, pedestrians, bicycles and motorcycles

Image classification



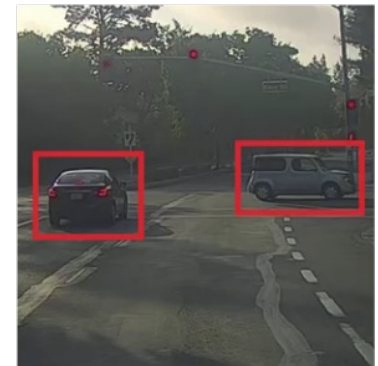
"Car"

Classification with localization



"Car"

Detection

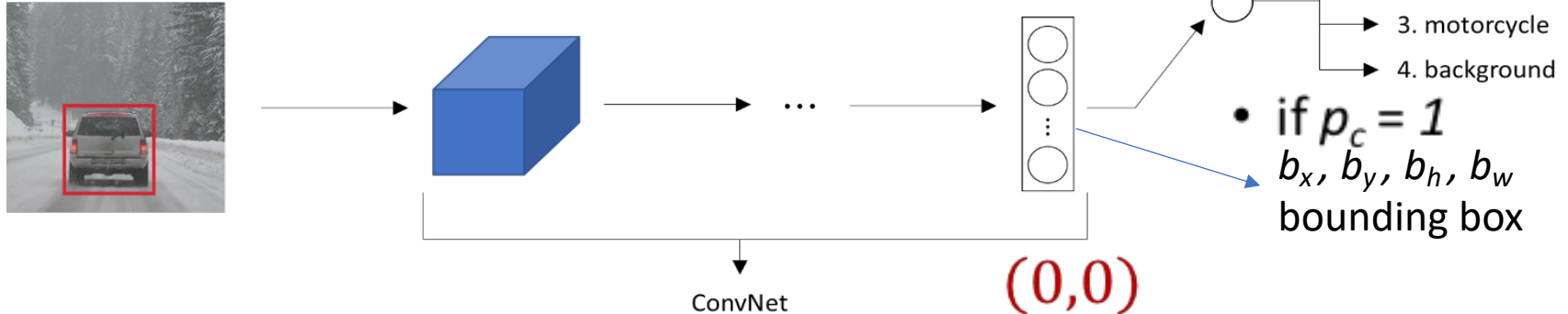


Multiple objects

One object

CNN Localization and Classification

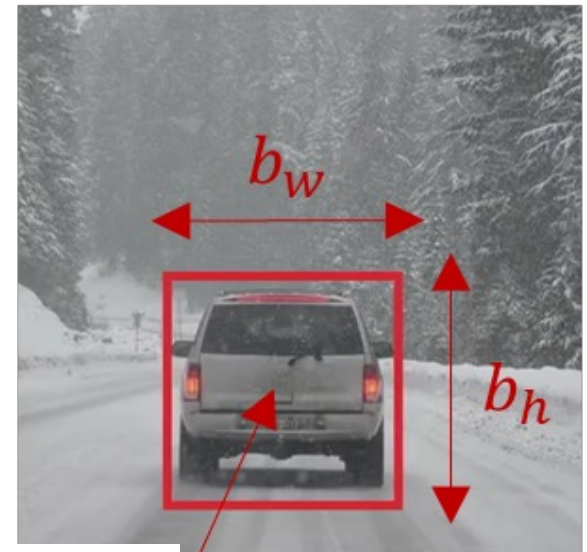
Assume a NN for classifying objects
(pedestrian, car, motorcycle, background)



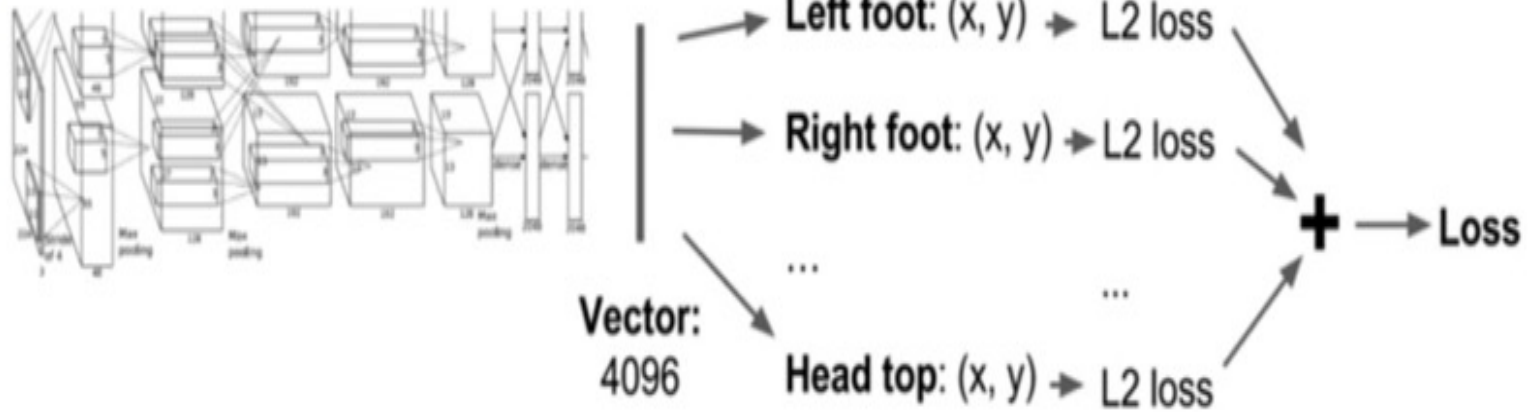
Four numbers, b_x, b_y, b_h, b_w specify the bounding box.

Example: $b_x = 0.5$; $b_y = 0.7$; $b_h = 0.3$; $b_w = 0.4$

- Output: $(p_c, b_x, b_y, b_h, b_w, c_1, c_2, c_3)$
where *if no object (background), $p_c = 0$*
if $p_c = 1$, then one of $c_1, c_2, c_3 = 1$
- Training set needs bounding boxes information and labels.
- Loss: $\lambda L_1 + L_2$ (L_1 =classification error; L_2 = bounding box error)



Pose Detection



$$\begin{array}{c}
 l_{1x}, l_{1y}, \\
 \vdots \\
 l_{16x}, l_{16y}
 \end{array}$$

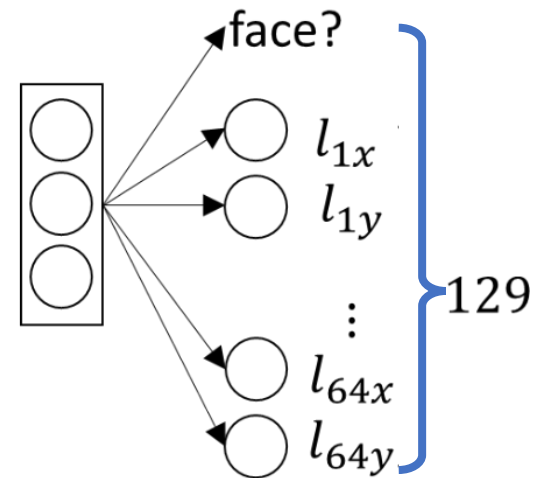
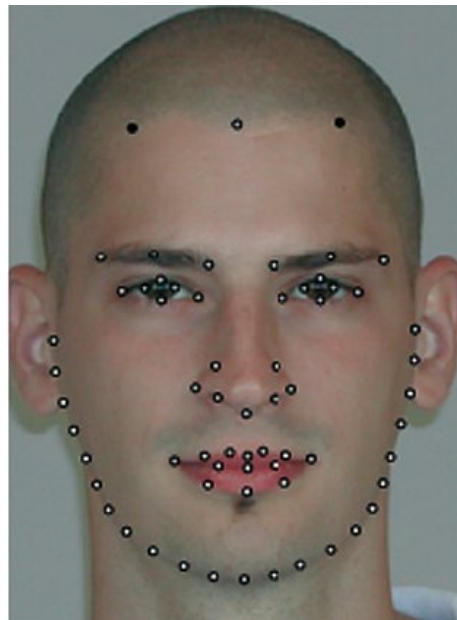
Landmark Detection

- Landmarks are coordinates of important points, e.g., Face recognition, positions of eyes



$$\left. \begin{array}{l} l_{1x}, l_{1y}, \\ l_{2x}, l_{2y}, \\ l_{3x}, l_{3y}, \\ l_{4x}, l_{4y}, \\ \vdots \\ l_{64x}, l_{64y} \end{array} \right\} x, y$$

→ ConvNet →



Various Computer Vision Tasks

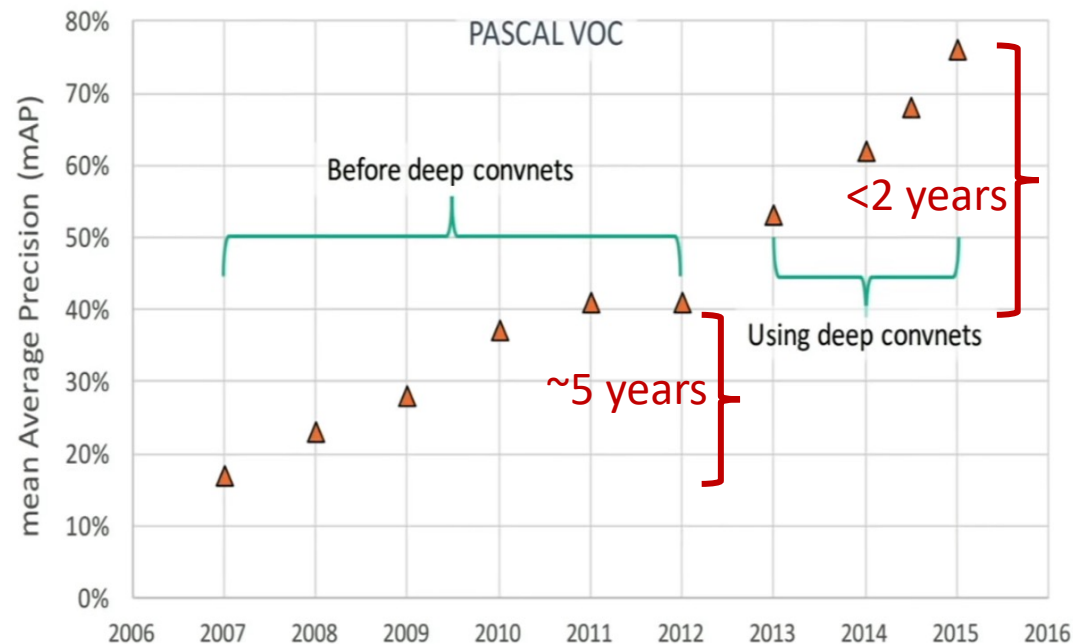
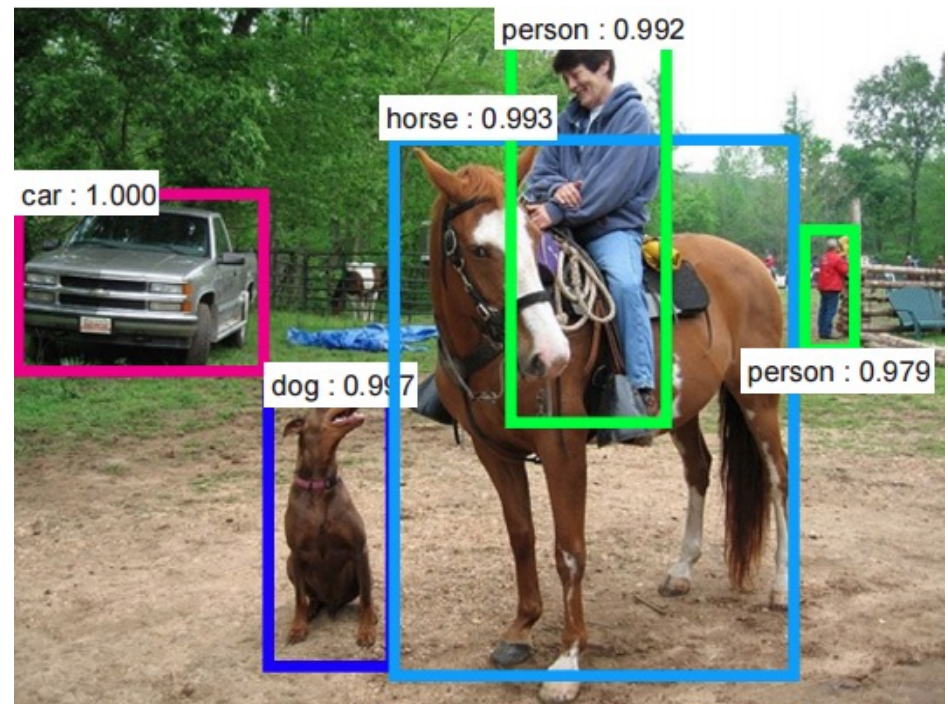
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Object Detection

Draw a bounding box and classify each object (unknown no. of objects)

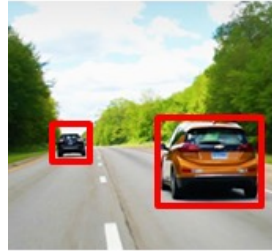
Solution: take different **regions of interest (RoI)** and use CNN to classify presence of object within each region.

Performance improves with Deep Learning



Object Detection

First create a labeled training set with cropped images



training set:



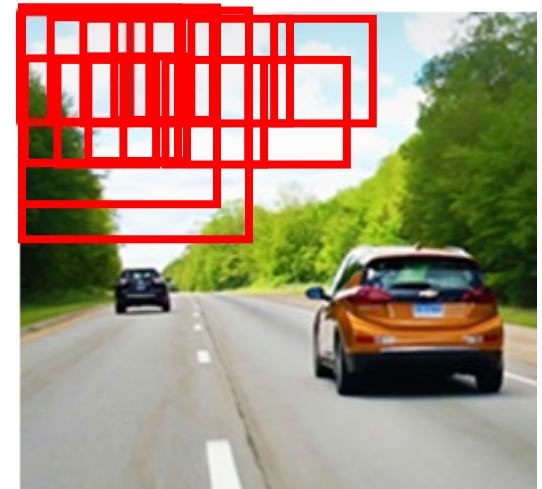
→ ConvNet → y

Sliding window detection

- Start with a little square at the top-right corner and check whether the little square contains a car
- Repeat by sliding the red square across the whole image
- Repeat with different sizes of square

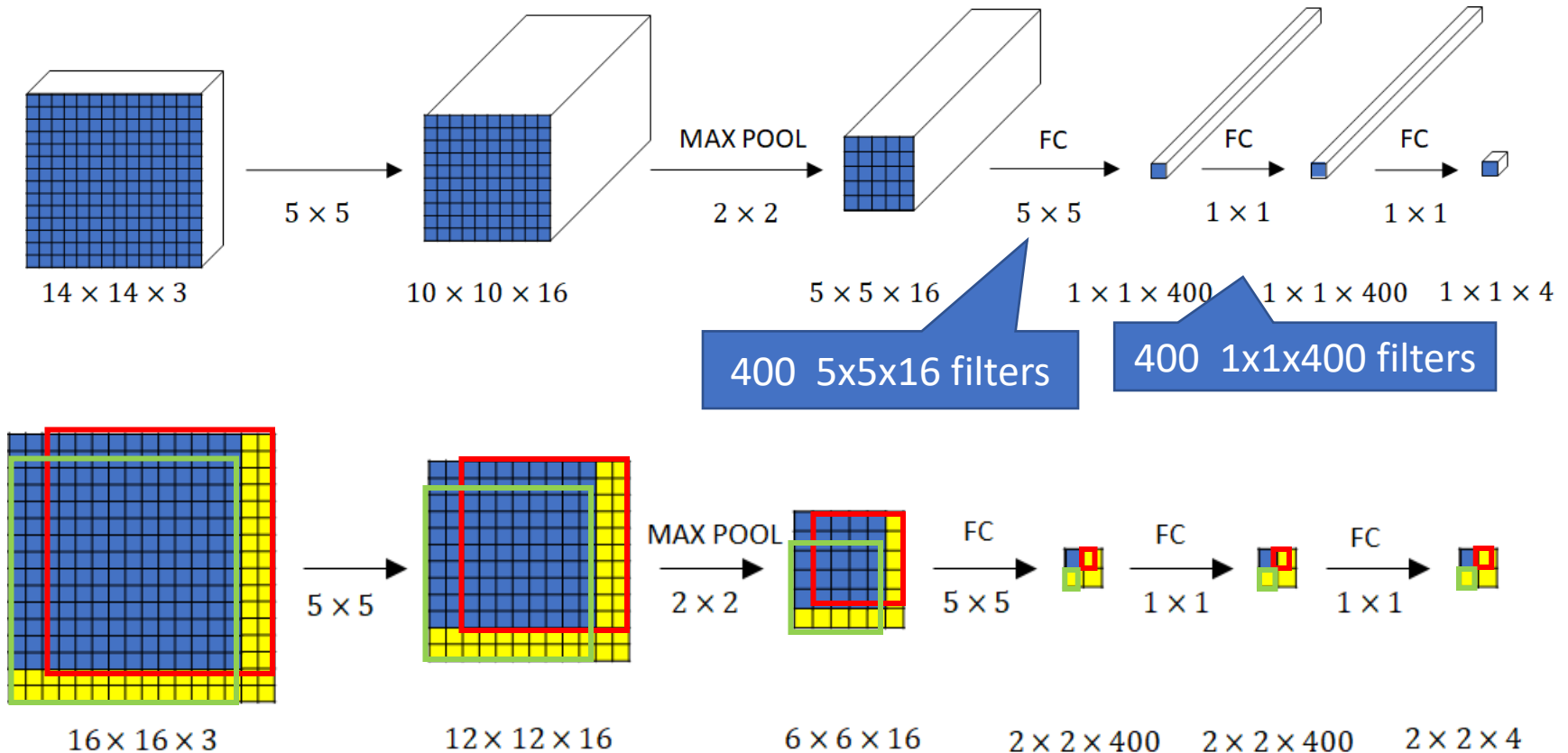
Disadvantage: huge computational cost

Too big stride or too big step size reduce computational cost, but hurt performance, i.e., miss the car.



Efficient Sliding Windows Computation

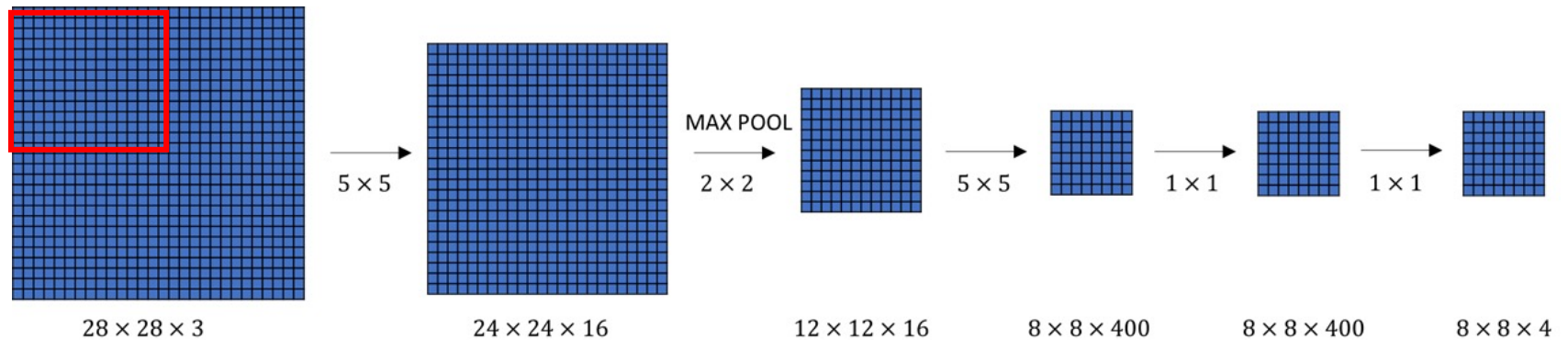
- *Fully connected layers* → *Convolutional layers*.



Combining all 4 propagations on 4 subsets of the input image into 1 convolution computation

Convolutional Operation of Sliding Windows

Convolutional implementation on a 28×28 image



Equivalent to sliding 14×14 windows (stride 2) on a 28×28 image.



End up with an $8 \times 8 \times 4$ output

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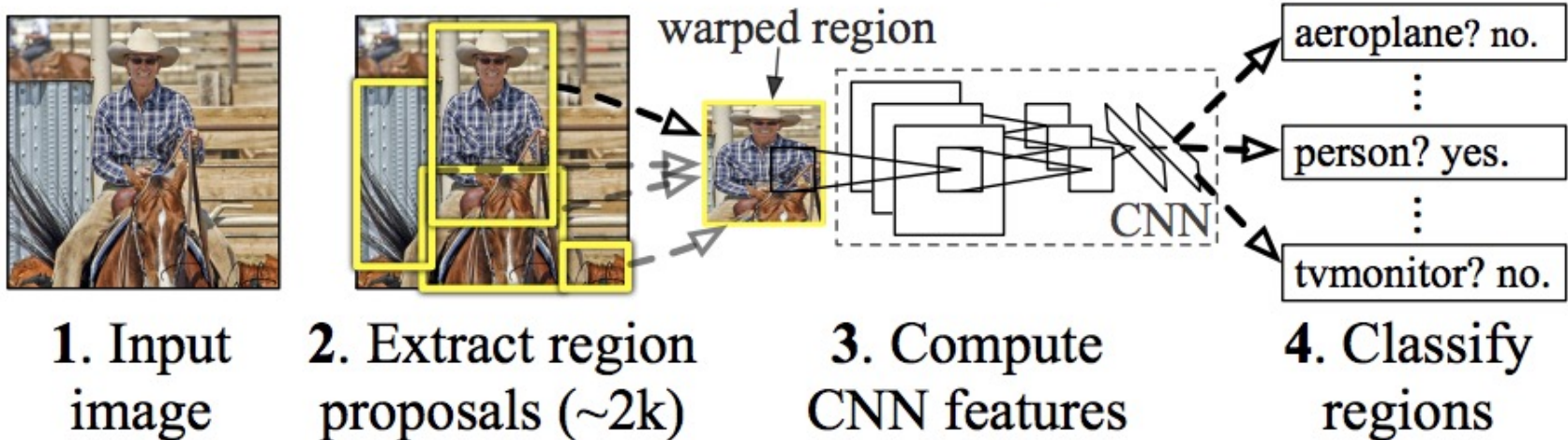
R-CNN (Region Based CNN)- 2014

Microsoft Research,
later join Facebook

Given an image

Girshick, et.al., "Rich Feature Hierarchies for Accurate Object Detection and Semantic Segmentation." 2014

- Find regions that might contain an object (*region proposals*)
- Extract CNN features from region proposals.
- Classify objects using extracted features.



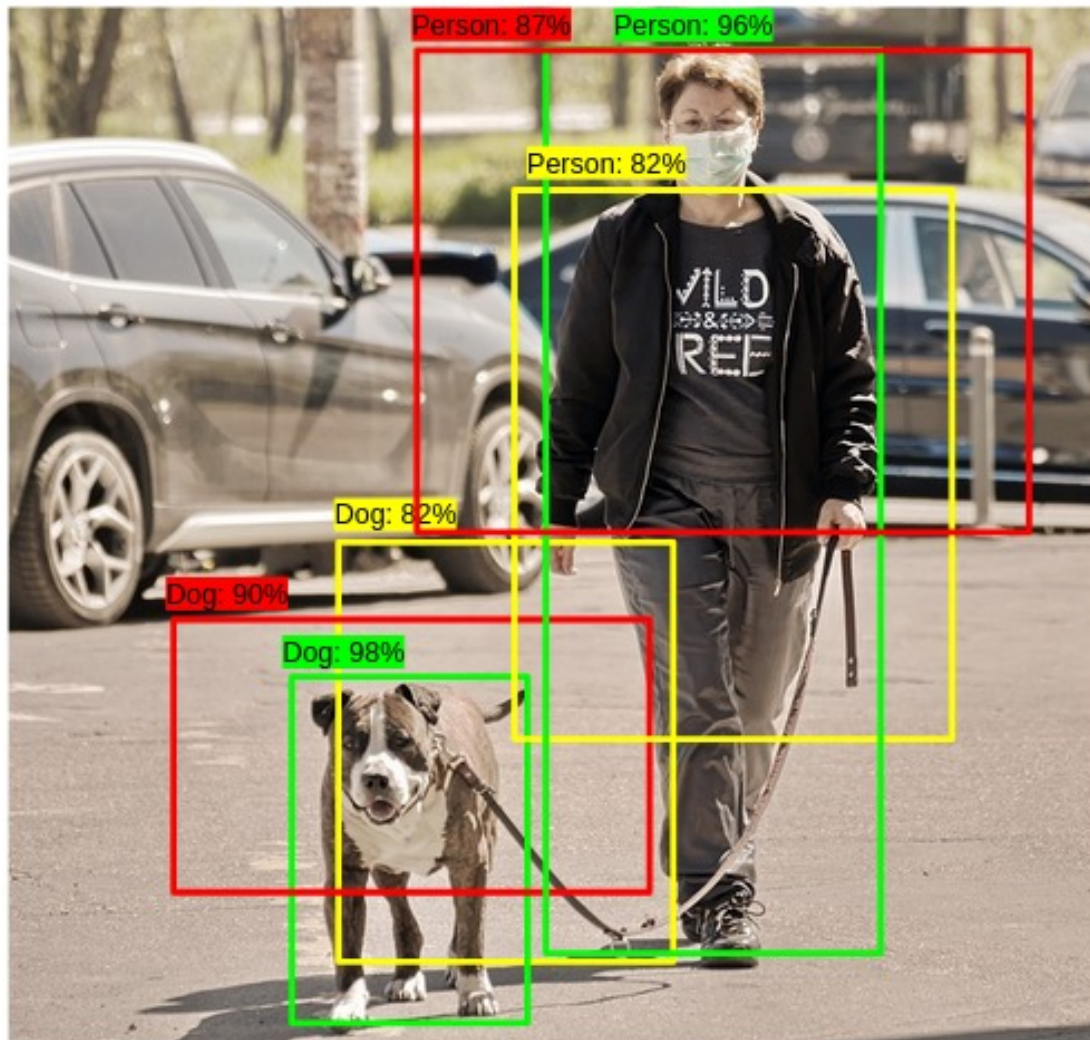
Selective search algorithm: generate 2000 region proposals

- Fixed algorithm (no learning feature) based on color, texture similarity and edges (# in box - # on box's boundary)
- For a training set of 1K images, 2M forward passes in training

Zitnick, et.al. "Edge boxes: Locating object proposals from edges." 2014.

Non-Max Suppression (NMS)

- Objects are of different sizes and shapes, detection algorithms create multiple bounding boxes.
- Non-max suppression selects the best bounding box (with highest objectiveness score) and rejects all others with high overlap.



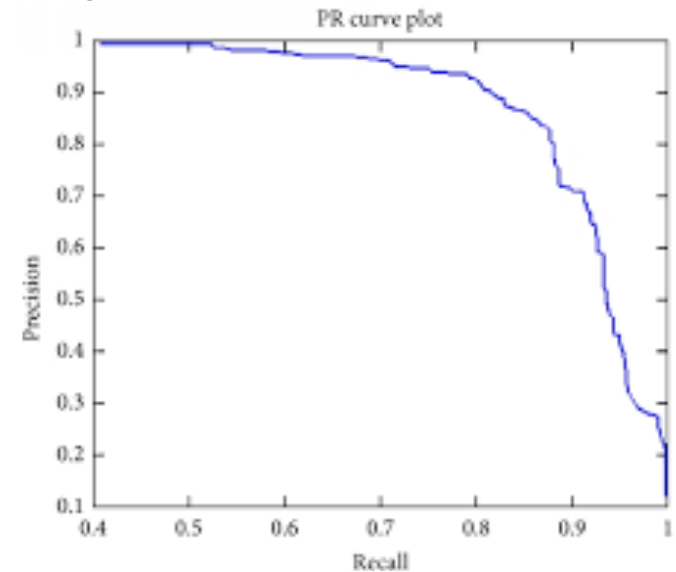
Performance evaluation for object detection

Average Precision (AP) –

area under a precision vs. recall curve by varying classification threshold for prediction

mAP –

AP average taken over all classes



Intersection over Union (IoU) – accuracy and errors between predicted and ground truth bounding boxes

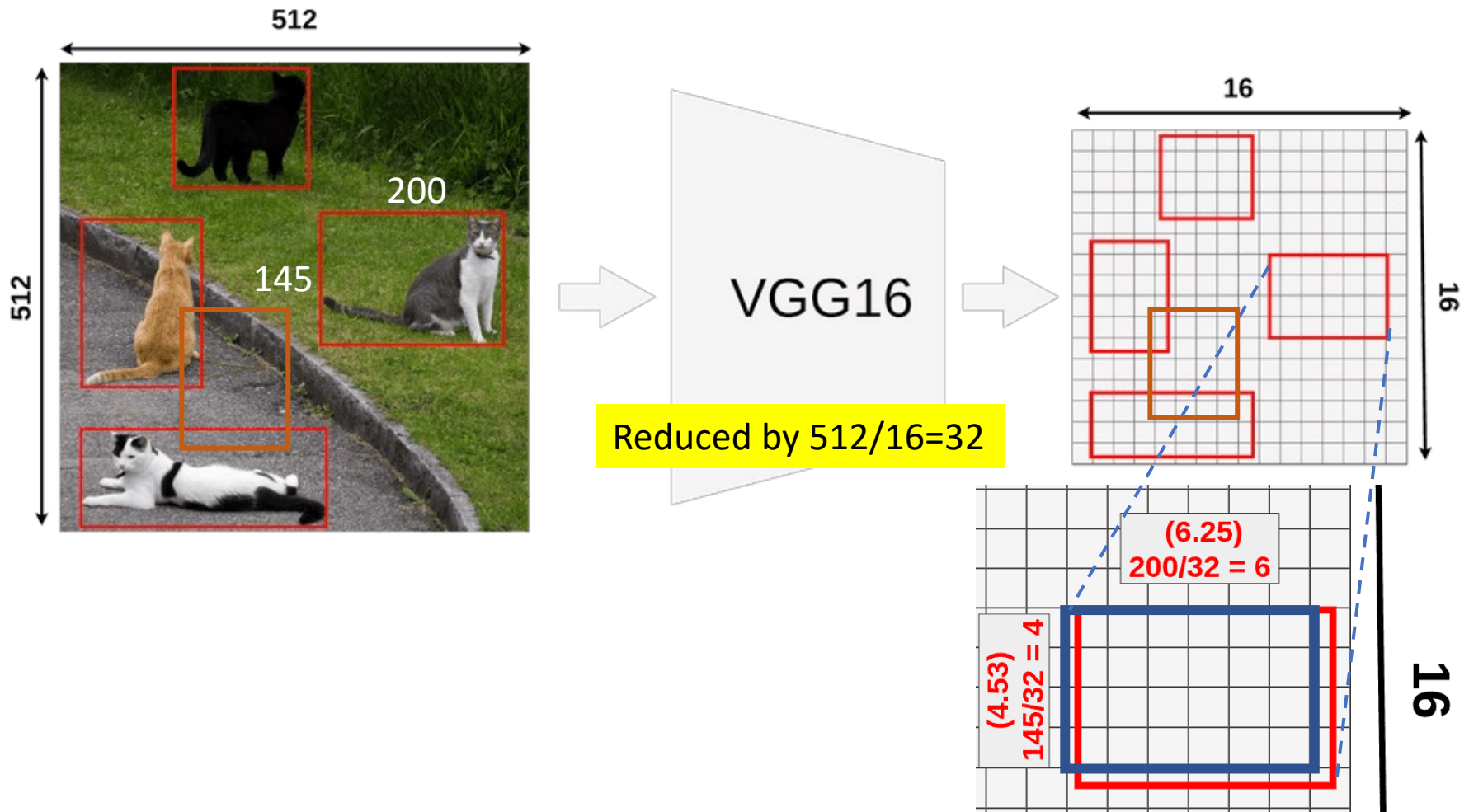
$$IoU = \frac{\text{Area of Overlap}}{\text{Area of Union}} =$$



Fast R-CNN (2015)

Girshick, Ross. Fast r-cnn." *IEEE ICCV*. 2015.

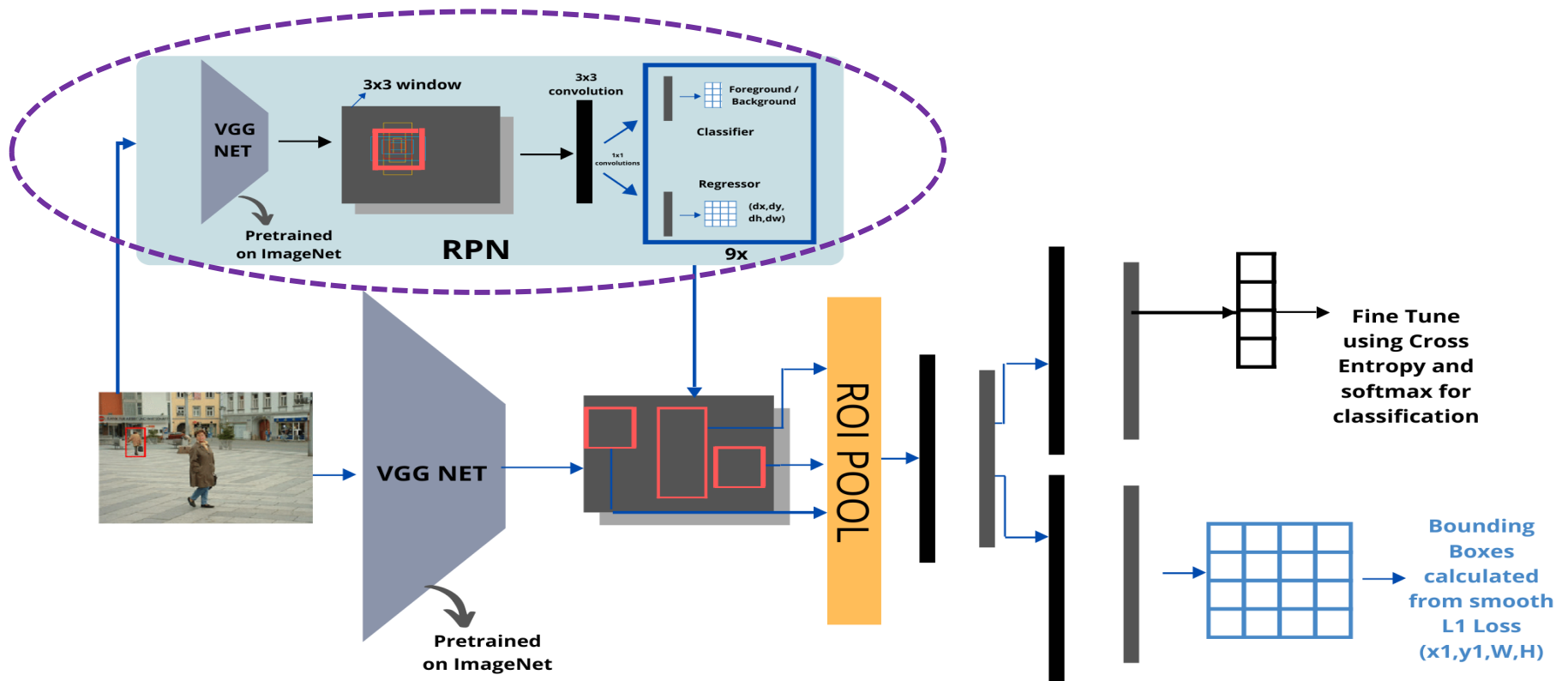
- The input image generates a conv feature map.
- Take region proposals as R-CNN before (external)
- Identify regions of proposals in the conv feature map



Faster R-CNN (2016)

Shaoqing Ren, et al, Faster R-CNN: Towards Real-Time Object Detection with Region Proposal Networks, 2016

- **Region Proposal Network** infers region proposals (not external) on top of Fast R-CNN



One- and Two-stage Object Detection

Traditional methods: two-stage object detector (2016)

- Stage 1 outputs region proposals (Region Proposal network), i.e., foreground vs background classification
- Stage 2 classifies each proposal (which class?)

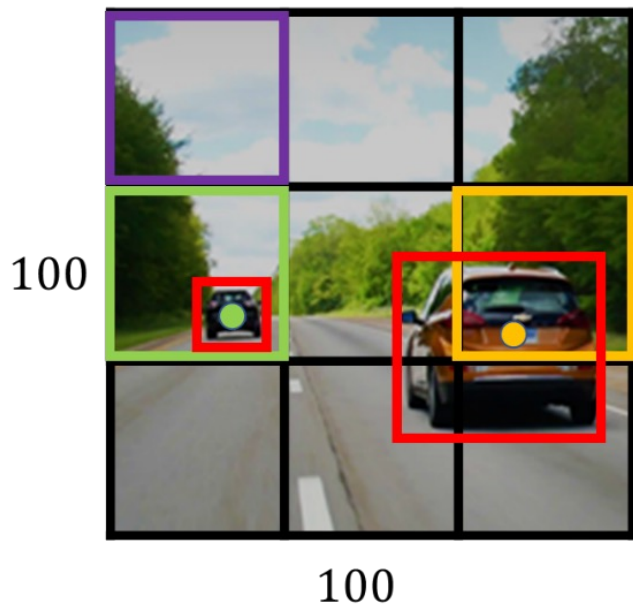
A **single shot detector** (SSD) outputs objects (classes) present in the image and their corresponding bounding box co-ordinates (multi-class classification and background) in a single pass. Example: YOLO, RetinaNet

Single-shot object detection is better suited for real-time applications, such as self-driving cars, video surveillance, ... while two-shot object detection is better for applications where accuracy is more important.

Bounding Box Predictions

Redmon, et al., **You Only Look Once**: Unified, Real-Time Object Detection, 2016

- **YOLO**: Partition the image into $S \times S$ grid (3x3 or 19x19), no multiple object per cell.
- Predicts multiple bounding boxes per cell and output the most accurate bounding boxes (single-stage). Each object will be assigned to just one grid cell



$$y = \begin{bmatrix} p_c \\ b_x \\ b_y \\ b_h \\ b_w \\ c_1 \\ c_2 \\ c_3 \end{bmatrix}$$

$\begin{bmatrix} 0 \\ ? \\ ? \\ ? \\ ? \\ ? \\ ? \\ ? \end{bmatrix}$
 $\begin{bmatrix} 1 \\ b_x \\ b_y \\ b_h \\ b_w \\ 0 \\ 1 \\ 0 \end{bmatrix}$
 $\begin{bmatrix} 1 \\ b_x \\ b_y \\ b_h \\ b_w \\ 0 \\ 1 \\ 0 \end{bmatrix}$
 → bounding box

$b_x = 0.3$
 $b_y = 0.9$ within (0, 1)
 $b_w = 1.1$
 $b_h = 1$ could be >1

Output size: 3x3x8

RetinaNet: apply Focal Loss to addresses class imbalance problem

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Applications (Segmentation)

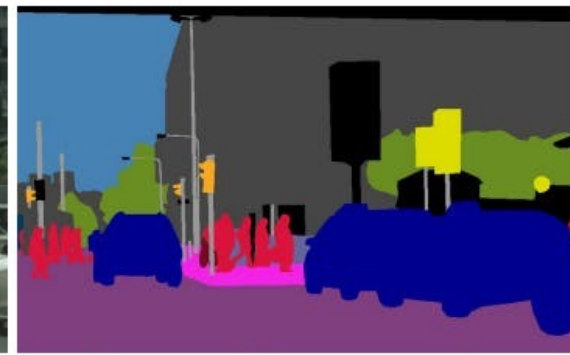
Scene understanding:
to detect drivable regions

Different from detection:

- no bounding boxes
- Do not distinguish different instances of the same object (**instance segmentation**)
e.g., same label for cars.



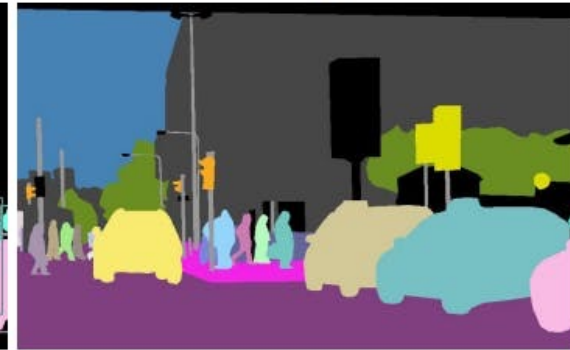
(a) image



(b) semantic segmentation

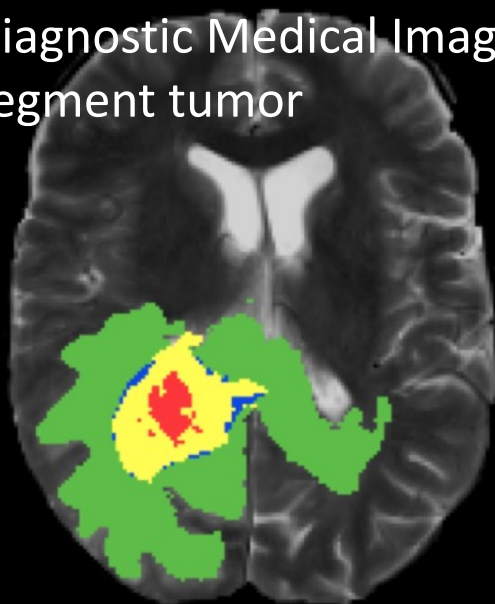


(c) instance segmentation

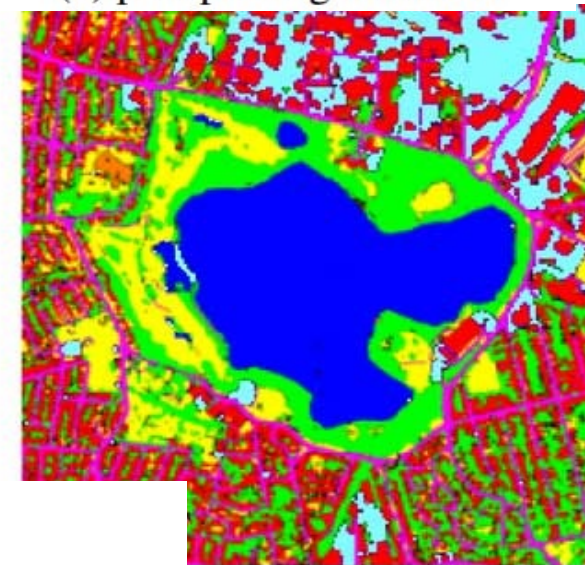


(d) panoptic segmentation

Diagnostic Medical Image
segment tumor

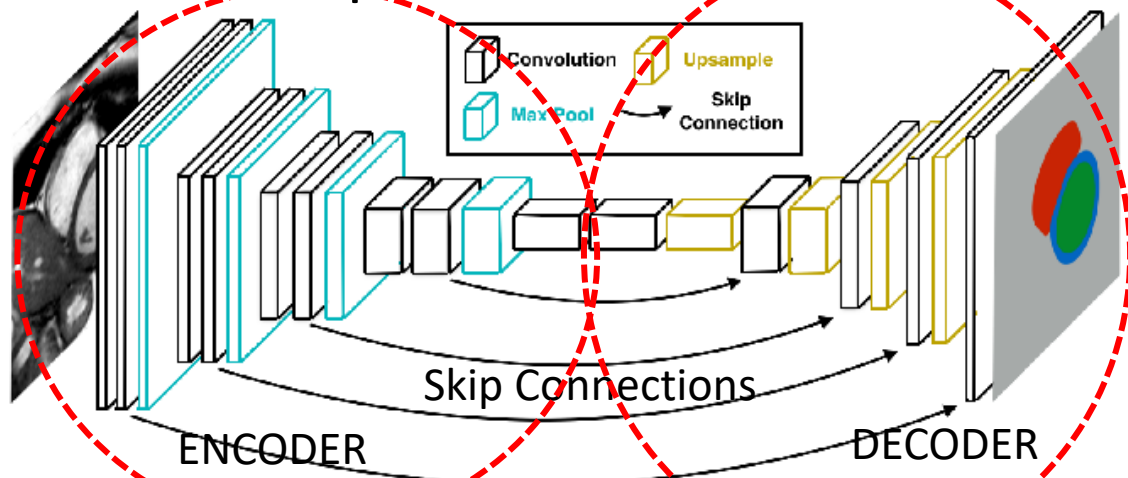


Satellite images
segment different types of land



Segmentation - Encoder-Decoder Structure (U-Net)

- **Encoder** output contains object information, shape and size
- **Decoder** decodes low-resolution image to produce high-resolution segmentation maps
 - adds more convolution layers coupled with upsampling layers
 - decreases no. of channels to retain low-level information
 - recovers lost information (inaccurate boundary).
 - accesses encoder's low-level features by **skip connections**.
- Skip Connection - Intermediate outputs of encoder are added/concatenated with the inputs to the intermediate layers of the decoder
 - uses the contextual feature information collected in encoder blocks to generate our segmentation map.



- U-Net won the ISBI (IEEE International Symposium on Biomedical Imaging) cell tracking challenge 2015 by a large margin.

